

VISUAL ARCHITECTURE FIELD GUIDE

Twelve reusable system diagrams plus decision prompts for lectures, downloads, slides, and capstone reviews.

OUTCOMES

SLOs - safety - cost - user value

AI WORKLOAD

training - inference - retrieval - agents

PLATFORM

pipelines - serving - orchestration - registry

RUNTIME

frameworks - containers - drivers - libraries

FOUNDATION

compute - network - storage - facilities

52 WEEKS	365 NUMBERED TOPICS	ZERO TO HERO
12 vector diagrams	Professional resource edition	Concepts + practice + evidence

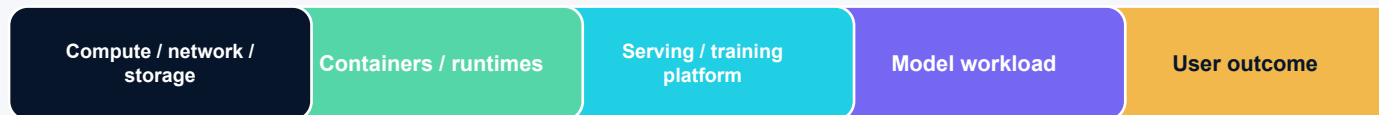
READ DIAGRAMS AS DECISIONS

A useful architecture diagram shows system boundaries, flow direction, interfaces, ownership, and failure domains. These visuals are deliberately vendor-neutral so they can be annotated during lectures or adapted to different clouds and tools.

For every arrow, ask: what crosses this boundary, how is it authenticated, what is the timeout, what is measured, and what happens when the destination is slow or unavailable?



Ai Infrastructure Stack



Begin at the outcome and trace constraints downward; trace evidence and failure signals upward.

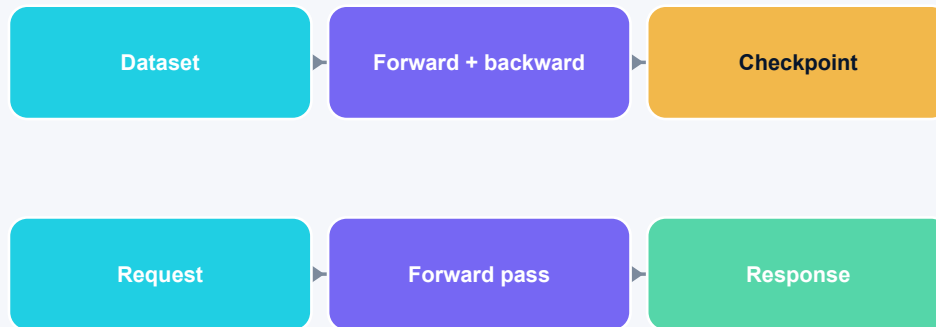
ANNOTATE

Boundary - interface - owner - trust - SLO - failure - metric - cost

REVIEW

Which assumption is most fragile? Which measurement would change the design?

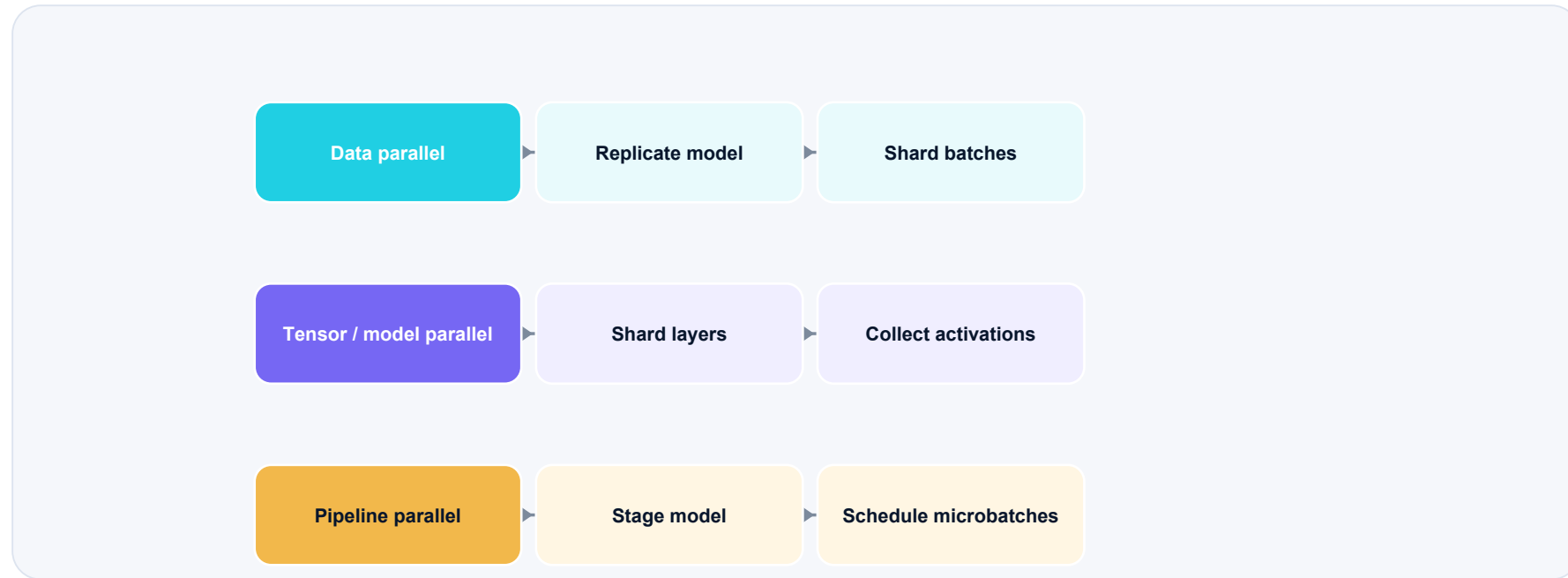
Training Vs Inference



Training optimizes model state; inference optimizes dependable response delivery.

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Distributed Parallelism



Parallelism is a memory and communication decision, not simply a device-count setting.

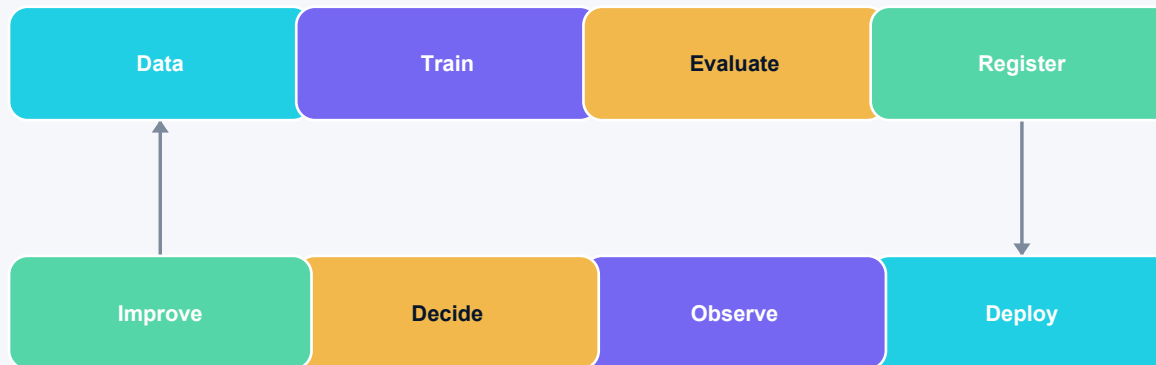
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Mlops Lifecycle



Lineage and evidence connect every lifecycle transition.

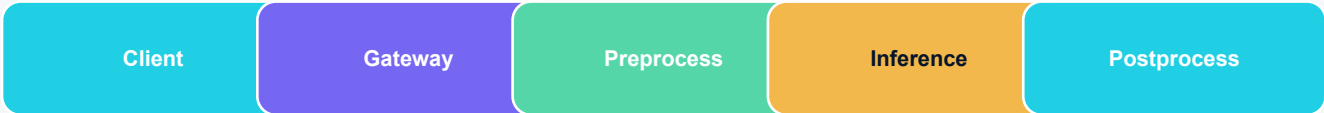
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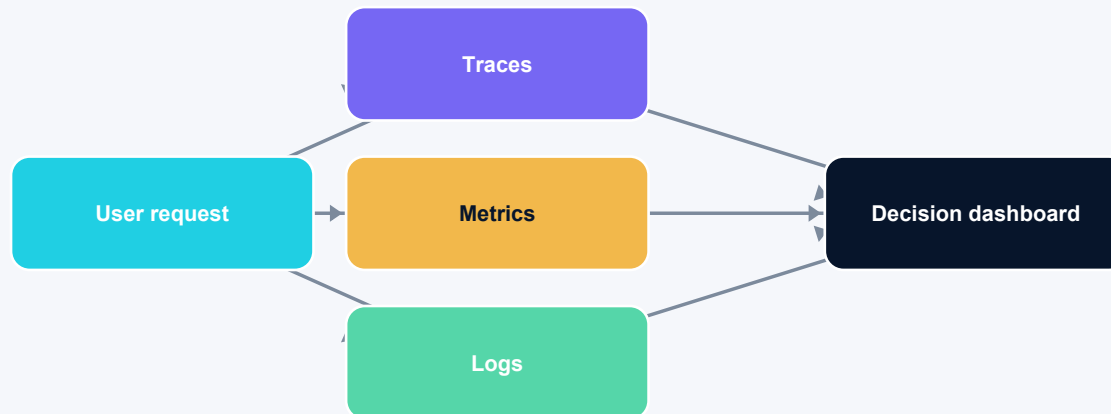
Model Serving Path



Assign latency, timeout, schema, ownership, and telemetry to every hop.

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Observability Signals



Telemetry is useful when it shortens a decision or recovery path.

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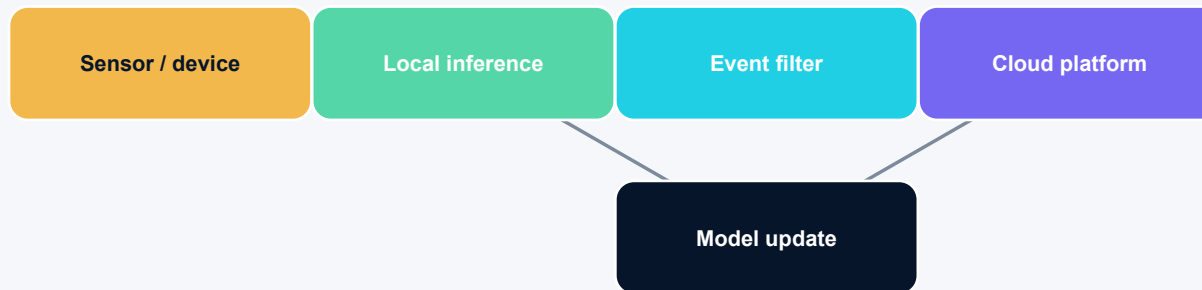
Rag Pipeline



Evaluate retrieval and generation separately before tuning the full system.

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Edge To Cloud



Keep urgent, private, or disconnected decisions local; centralize learning and fleet control where justified.

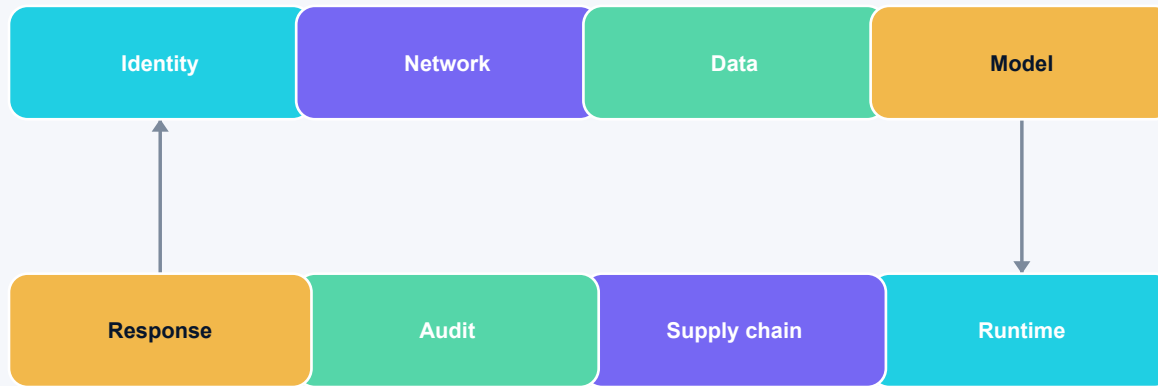
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Security Defense In Depth



Controls should prevent, detect, contain, and recover across the entire asset path.

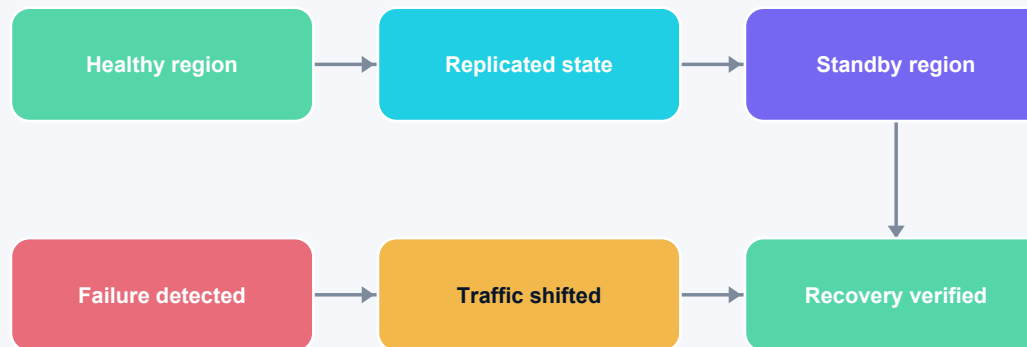
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Ha And Disaster Recovery



Availability protects service continuity; disaster recovery restores capability within RTO and RPO.

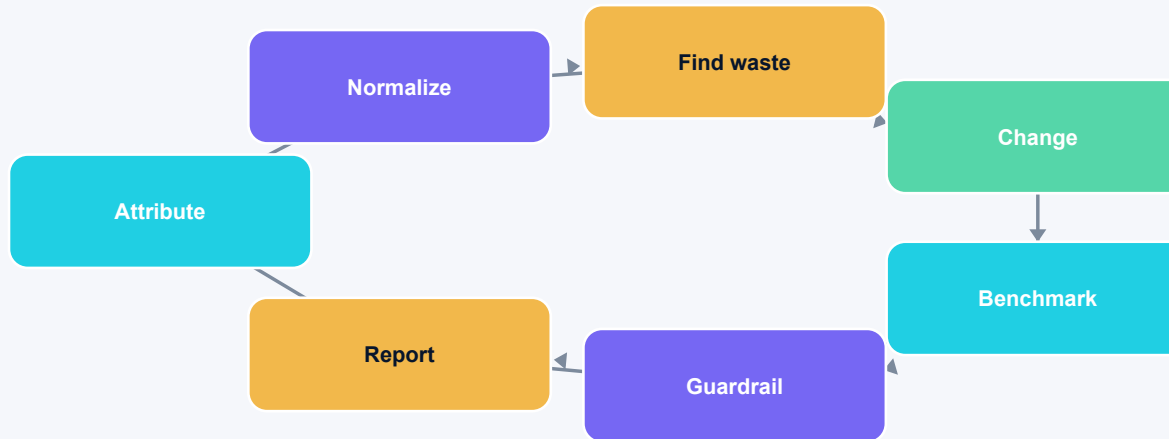
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Cost Optimization Loop



Optimize unit cost while protecting quality, reliability, security, and engineering velocity.

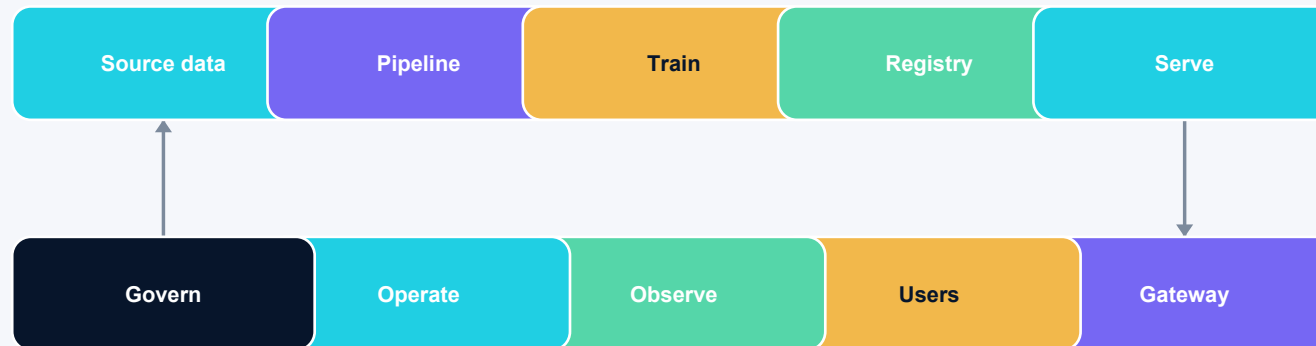
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Capstone Reference Architecture



A defensible capstone connects every component to a requirement, interface, metric, failure mode, and owner.

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CAPSTONE DIAGRAM REVIEW

Before presenting, make sure the architecture visual can answer these questions without verbal rescue.

- What user or business outcome is the system responsible for?
- Where do data, model artifacts, configuration, and secrets originate and persist?
- Which paths are synchronous and latency-sensitive? Which are batch or asynchronous?
- Where are trust boundaries, identities, permissions, encryption, and audit records?
- Which components scale independently, retain state, or create a single point of failure?
- Which signals prove quality, latency, throughput, reliability, security, and unit cost?
- How does the system roll back, fail over, restore, and safely degrade?
- What is deliberately omitted, and what evidence justifies the tradeoff?